Supply Chain SLA Diagnostic & Fulfillment Bottleneck Analysis

STRATEGIC DILEMMA

The Core Business Inquiry

> **Where in the fulfillment lifecycle are delivery delays originating—seller dispatch or courier transit—and which corridors systematically breach SLAs?**

First-Mile Check

Detecting merchant packaging delays beyond shipping limit dates.

Transit Risk Check

Auditing last-mile.

BASELINE PERFORMANCE

Executive KPI Scorecard

Diagnostics across 110.2K shippings

110.2K	7.91%	9.31%	5.66%	9.19
Total Shipments	Customer SLA Breach	Seller Dispatch Breach	Carrier Failure Rate	Avg Transit Days

Operational Benchmark: Across the 8,722 delayed orders, road transit times dominate platform friction.

ROOT-CAUSE

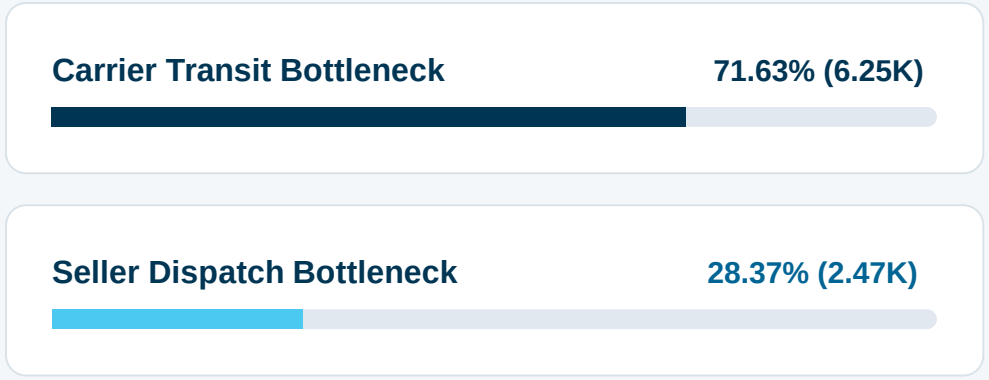
Delays Originate in Transit

Decomposing 8,722 delayed shipments into operational failure nodes

71.6%

Carrier Transit Failures

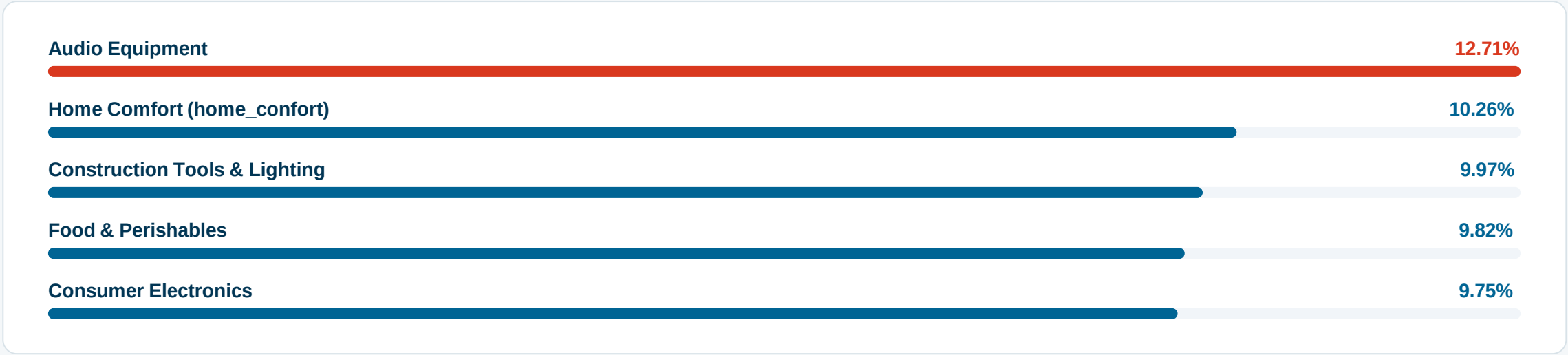
6,248 delayed parcels met merchant handoff deadlines but failed in road transit.



Primary operational takeaway: Courier performance drives 2.5x more late deliveries than merchant prep delays.

Top High-Risk Product Categories

Filtered for categories with ≥ 300 shipments (~95.3% volume coverage)



GEOGRAPHIC ANALYSIS

The Interstate Delivery Penalty

Cross-state logistics drive a 50.5% surge in customer SLA breaches

CROSS-STATE

9.00%

Inter-State Breach Rate

SAME-STATE

5.98%

Intra-State Breach Rate

1.5x Performance Disparity

Passing state borders introduces late deliveries

Transit Time Impact

Interstate routes average 7 additional days compared to intrastate routes.

PHYSICAL CONSTRAINTS

Weight Tiers & Heavy Cargo Friction

Assessing parcel mass constraints against courier delivery velocity

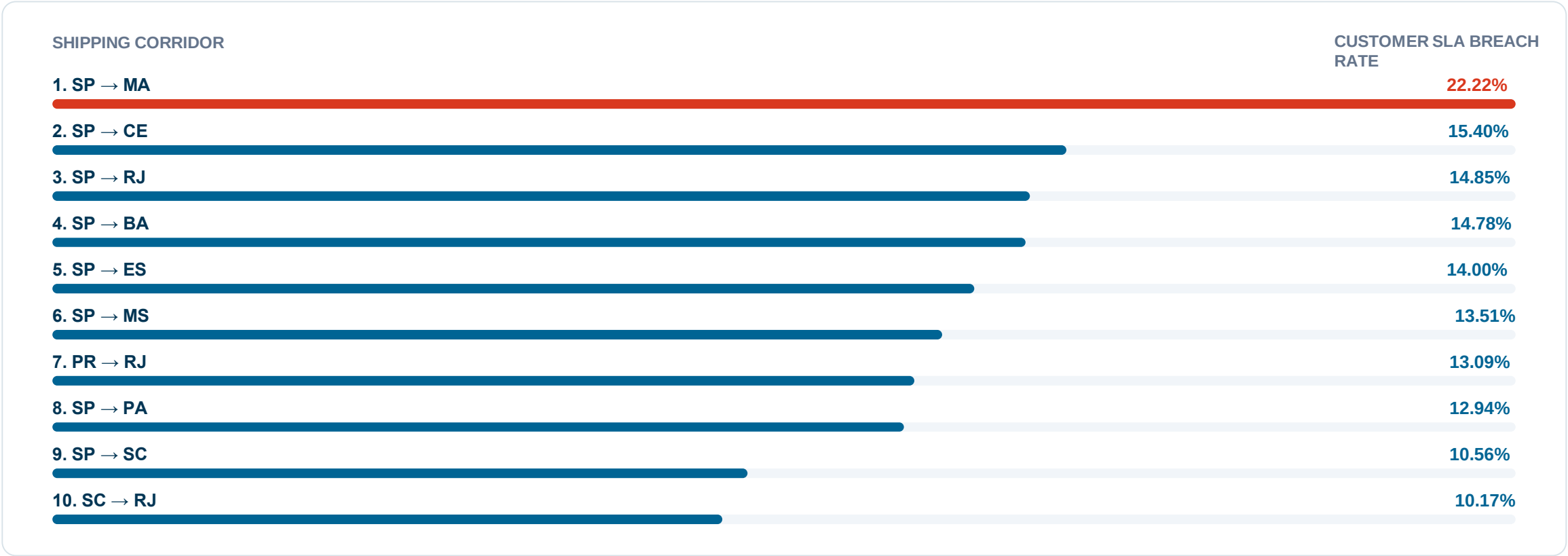
Under 1 kg	1 kg - 5 kg	5 kg - 10 kg	Over 10 kg
7.53%	8.28%	7.77%	10.61%
High Volume (~60K)	Standard weight	Medium Packages	+40.9% Risk Spike

Handling Bottleneck: Packages over 10 kg jump to a 10.61% breach rate (a 40.9% increase over packages under 1 kg at 7.53%)..

LANE DIAGNOSTICS

Top 10 Worst Shipping Routes

Filtered for routes having ≥ 500 shipments (covers 86.2% of platform volume)



Volume Threshold Integrity

Preventing small-sample distortions and ranking false alarms

Categories (≥ 300 Orders)

95.3% Coverage

Retains 35 core categories out of 73. Drops 38 long-tail niche categories (under 5% platform volume) to eliminate false alarms.

Corridors (≥ 500 Shipments)

86.2% Coverage

Isolates 30 main routes from 400+ corridors. Guarantees daily dispatch frequency (~1 dispatch/day) to test high volume courier routes.

STRATEGIC ROADMAP

Operational Action Plan

Dynamic Lead Times

Add 3–4 day checkout buffer.

Dedicated Bulky Orders Courier

Contract specialized carriers for parcels exceeding 10 kg.

Regional Warehousing

Position more volume categories in regional hubs.

Carrier SLA Penalties

Enforce penalty clawbacks on lanes exceeding 8% breach rates.

Thank You

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